

Supplementary Text S1

List of Parameters

	Parameter	Description	Units	Value	Reference [#]
1.	β_{M2}	Rate of IL-10 production from M2 cells	ng cell ⁻¹ day ⁻¹	1x10 ⁻¹⁵	Expected
2.	β_{Tc}	Rate of IFN- γ production from Tc cells	ng cell ⁻¹ day ⁻¹	1x10 ⁻⁸	Expected
3.	β_{Th1CK2}	Rate of IFN- γ production from T _{H1} cells	ng cell ⁻¹ day ⁻¹	1x10 ⁻⁷	Expected
4.	β_{Th1CK3}	Rate of IL-2 production from T _{H1} cells	ng cell ⁻¹ day ⁻¹	1 x10 ⁻⁸	Expected
5.	β_{Th2}	Rate of IL-10 production from T _{H2} cells	ng cell ⁻¹ day ⁻¹	1 x10 ⁻⁹	Expected
6.	β_{Treg}	Rate of IL-10 production from Treg cells	ng cell ⁻¹ day ⁻¹	1 x10 ⁻¹⁰	Expected
7.	γ_C	Birth rate of Cancer cells	day ⁻¹	0.1282	Estimated
8.	γ_{CR}	Birth rate of Cancer resistant cells	day ⁻¹	0.1282	Expected
9.	γ_{M1}	Birth rate of M1 cells	day ⁻¹	0.7	Expected
10.	γ_{M2}	Birth rate of M2 cells	day ⁻¹	0.01	Expected
11.	γ_S	Birth rate of Stem cells	day ⁻¹	0.15	Expected
12.	γ_{Tc}	Birth rate of Tc cells	day ⁻¹	1.0	[1]
13.	γ_{Th1}	Birth rate of T _{H1} cells	day ⁻¹	2.0	[2]
14.	γ_{Th2}	Birth rate of T _{H2} cells	day ⁻¹	2.0	[2]
15.	γ_{Treg}	Birth rate of Treg cells	day ⁻¹	0.3	[3]
16.	δ_C	Death rate of Cancer cells	day ⁻¹	0.8055	Estimated
17.	δ_{Ck1}	Degradation rate of IL-10	day ⁻¹	19.757	Estimated
18.	δ_{Ck2}	Degradation rate of IFN- γ	day ⁻¹	6.1212	Estimated
19.	δ_{Ck3}	Degradation rate of IL-2	day ⁻¹	8.664339	Calculated [4]
20.	δ_{CR}	Death rate of Resistant Cancer cells	day ⁻¹	5.37 x10 ⁻⁵	Estimated
21.	δ_{M1}	Death rate of M1 cells	day ⁻¹	1.02	[5]
22.	δ_{M2}	Death rate of M2 cells	day ⁻¹	0.05	[5]
23.	δ_S	Death rate of Stem cells	day ⁻¹	2 x10 ⁻⁷	Expected
24.	δ_{Tc}	Death rate of Tc cells	day ⁻¹	5.2939	Estimated
25.	δ_{Th1}	Death rate of T _{H1} cells	day ⁻¹	2.0	[6]
26.	δ_{Th2}	Death rate of T _{H2} cells	day ⁻¹	2.0	Expected
27.	δ_{Treg}	Death rate of Treg cells	day ⁻¹	1.0	[7]
28.	λ_{M1}	Saturation constant for M1 activation	cells ml ⁻¹	1x10 ⁸	Expected
29.	λ_{M2}	Saturation constant for M2 activation	cells ml ⁻¹	1 x10 ⁶	Expected

30.	λ_{Tc1}	Saturation constant for Tc activation by cancer cells	cells ml ⁻¹	1 x10 ⁵	Expected
31.	λ_{Tc2}	Saturation constant for Tc inhibition by stem cells	cells ml ⁻¹	5 x10 ⁵	Expected
32.	λ_{Tc3}	Saturation constant for Tc inhibition by Treg cells	cells ml ⁻¹	5 x10 ¹⁰	Expected
33.	λ_{Tc4}	Saturation constant for Tc activation by T _{H1} cells	cells ml ⁻¹	1 x10 ⁵	Expected
34.	λ_{Th1}	Saturation constant for T _{H1}	cells ml ⁻¹	1 x10 ⁵	Expected
35.	λ_{Th2}	Saturation constant for T _{H2}	cells ml ⁻¹	1 x10 ⁵	Expected
36.	λ_{Treg2}	Saturation constant for Treg	cells ml ⁻¹	1 x10 ⁷	Expected
37.	μ_{C1}	Rate of activation of C and C _R by IL-10	day ⁻¹	0.75	[8]
38.	μ_{C2}	Rate of killing of C and C _R by IFN- γ	day ⁻¹	0.9	[9]
39.	μ_S	Rate of killing of S by IFN-gamma	day ⁻¹	0.17	[9]
40.	μ_{SR}	Rate of killing of S _R by IFN-gamma	day ⁻¹	0.18	Expected
41.	μ_{TcS}	Rate of Tc killing by S and S _R cells	day ⁻¹	1 x10 ⁻¹⁰	[10]
42.	μ_{TcTreg}	Rate of Tc killing by Treg cells	day ⁻¹	1.5x10 ⁻⁵	[11]
43.	μ_{Th1Ck1}	Rate of T _{H1} killing by IL-10	day ⁻¹	1 x10 ⁻⁹	[12]
44.	μ_{Th1Ck3}	Rate of T _{H1} activation by IL-2	day ⁻¹	0.1245	[13]
45.	$\mu_{TregCk1}$	Rate of Treg activation by IL-10	day ⁻¹	1 x10 ⁻⁷	Expected
46.	$C_{max}=K_{tumor}/2$	Carrying capacity for Cancer cells , where K _{tumor} =2 x10 ¹⁰ is the total Carrying capacity of non-stem tumor cells	cells ml ⁻¹	1 x10 ¹⁰	[14]
47.	$CR_{max}=K_{tumor}/2$	Carrying capacity for Resistant Cancer cells, where K _{tumor} =2 x10 ¹⁰ is the total Carrying capacity of non-stem tumor cells	cells ml ⁻¹	1 x10 ¹⁰	Expected
48.	k_1	Saturation constant for inhibition of S by IFN- γ	ng ml ⁻¹	10.0	Expected
49.	k_{11}	Saturation constant for activation of Treg by IL-10	ng ml ⁻¹	0.001	Expected
50.	k_2	Saturation constant for inhibition of S _R by IFN- γ	ng ml ⁻¹	10.0	Expected
51.	k_3	Saturation constant for activation of C by IL-10	ng ml ⁻¹	2.0531	Estimated
52.	k_4	Saturation constant for inhibition of C by IFN- γ	ng ml ⁻¹	3.02	Estimated
53.	k_5	Saturation constant for activation of C _R by IL-10	ng ml ⁻¹	6.7979	Estimated
54.	k_6	Saturation constant for inhibition of C _R by IFN- γ	ng ml ⁻¹	6.9937	Estimated

55.	k8	Saturation constant for inhibition of T_{HI} by IL-10	ng ml ⁻¹	0.01	Expected
56.	k9	Saturation constant for activation of T_{HI} by IL-2	ng ml ⁻¹	0.001	Expected
57.	k_{tc1}	Saturation constant for inhibition of S by Tc	cell ml ⁻¹	1 x10 ⁹	Expected
58.	k_{tc2}	Saturation constant for inhibition of S_R by Tc	cell ml ⁻¹	1 x10 ⁸	Expected
59.	k_{tc3}	Saturation constant for inhibition of C by Tc	cell ml ⁻¹	1 x10 ⁹	Expected
60.	k_{tc4}	Saturation constant for inhibition of C_R by Tc	cell ml ⁻¹	1 x10 ⁹	Expected
61.	m_C	Probability of Stem cell transformation into Stem resistant cells	-	0.01	Expected
62.	m_S	Probability of Cancer cell transformation into Cancer resistant cells	-	4 x10 ⁻⁷	[14]
63.	p1	Probability of Asymmetric differentiation of stem cells	-	0.2	[14]
64.	p2	Probability of Symmetric differentiation into two differentiated cancer cells	-	0.05	[14]
65.	r1	constant		0.0001	Expected
66.	r2	constant		1x x10 ⁻⁵	Expected
67.	tck	Rate of killing of tumor by Tc cells	day ⁻¹	0.1	Expected
68.	μ_{M1Ck2}	Rate of M1 activation by IFN-γ	day ⁻¹	0.01	Expected
69.	μ_{M2Ck1}	Rate of M2 activation by IL10	day ⁻¹	0.01	Expected
70.	k7	Saturation constant for proliferation of M1 by IFN-γ	ng ml ⁻¹	0.2	Expected
71.	k10	Saturation constant for proliferation of M2 by IL10	ng ml ⁻¹	0.01	Expected

Estimated parameter values have been determined by the MCMC techniques using the time course experiment cytometric data for cancer cell proliferation for Gastric cancer cell line (SGC7901) (Refer *Methods* of Main Article) [15].

Expected parameter values are estimated by varying the parameters within the biologically feasible ranges found in various Literatures so as to determine its expected value to calibrate the model with experimental observations.

List of Initial values

Variable	Symbol	Initial Values	Reference
Stem Cell	S	1	-
Stem Resistant Cell	S_R	0	-
Cancer Cell	C	0	-
Cancer Resistant Cell	C_R	0	-
Type-I Tumor Associated Macrophage	M1	85000	Calculated [5]
Type-II Tumor Associated Macrophage	M2	15000	Calculated [5]
Type-I Helper T Cell	T_{H1}	71000	[16]
Type-II Helper T Cell	T_{H2}	12000	[16]
Cytotoxic T Cell	T_c	56000	[16]
Regulatory T Cell	Treg	8000	[16]
Interleukin-10	IL10	0.0085	[17]
Interferon- γ	IFN- γ	0.12	[17]
Interleukin-2	IL2	0.0094	[17]

Reference

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